

Integration with Ln

Section 5.2

11/14/25

RULE A

$$\frac{d}{dx} \ln u = \frac{u'}{u}$$

$$\int \frac{1}{u} du = \ln|u| + C$$

$$\int 5x^2 + 3x + 4 + x^{-1} dx =$$

$$\frac{5}{3}x^3 + \frac{3}{2}x^2 + 4x + \ln|x| + C$$

$$\int \frac{1}{5x-3} dx \quad \text{vs} \quad \int \frac{1}{\sqrt{5x-3}} dx$$

$$u = 5x-3$$

$$du = 5dx$$

$$u = 5x-3$$

$$du = 5dx$$

$$\frac{1}{5} \int \frac{1}{u} du$$

$$\frac{1}{5} \int \frac{1}{\sqrt{u}} du$$

$$\frac{1}{5} \ln|u| + C$$

$$\frac{1}{5} (2(u)^{\frac{1}{2}}) + C$$

$$\frac{1}{5} \ln|5x-3| + C$$

$$\frac{1}{5} \ln(2(5x-3)^{\frac{1}{2}}) + C$$

$$\int \frac{x^2 + 5x + 4}{x} dx =$$

$$\int x + 5 + \frac{4}{x} dx =$$

$$\frac{1}{2}x^2 + 5x + 4 \ln|x| + C$$

$$\int \frac{e^{2x}}{1+e^{2x}} dx =$$

$$u = 1+e^{2x}$$

$$du = 2e^{2x}$$

$$\frac{1}{2} \int u^{-1} du =$$

$$\frac{1}{2} \ln|u| + C =$$

$$\frac{1}{2} \ln|1+e^{2x}| + C$$

$$\int \frac{(\ln x)^2}{x} dx$$

$$u = \ln x$$

$$du = \frac{1}{x} dx$$

$$\int u^2 du =$$

$$\frac{1}{3} u^3 + C$$

$$\frac{1}{3} (\ln x)^3 + C$$

$$\text{vs} \int \frac{1}{x \ln x} dx$$

$$u = \ln x$$

$$du = \frac{1}{x} dx$$

$$\int \frac{1}{u} du$$

$$\ln|u| + C$$

$$\ln|\ln x| + C$$

Rule B

When Degree on top >

Same or larger...

$$\int \frac{2x^2 + 7x - 13}{x-2} dx =$$

$$\begin{array}{r|rrrr} 2 & 2 & 7 & -13 & \\ & \downarrow & +4 & +22 & \\ \hline & 2 & 11 & 9 & \end{array}$$

$$\int 2x + 11 + \frac{9}{x-2} dx =$$

$$x^2 + 11x + 9 \ln|x-2| + C$$